

# GENERAL DATA (SERIES "TD")

|                                                  |                                                                                     |
|--------------------------------------------------|-------------------------------------------------------------------------------------|
| ENGINE TYPE                                      | XPAG/TD up to Engine No. 9407.<br>XPAG/TD/2 from Engine No. 9408.                   |
| Number of cylinders                              | Four.                                                                               |
| Capacity                                         | 1250 c.c. (76½ cu. in.).                                                            |
| B.H.P.                                           | 54.4 at 5,200 r.p.m.                                                                |
| Bore                                             | 66.5 mm. (2.618 in.).                                                               |
| Stroke                                           | 90 mm. (3.543 in.).                                                                 |
| R.A.C. rating                                    | 10.97 h.p.                                                                          |
| Compression ratio                                | 7.25 to 1.                                                                          |
| System of cooling                                | Thermo-siphon—pump and fan assisted.                                                |
| Radiator hose top                                | Length 4 in. (10.16 cm.).<br>Diameter 2⅝ in. (5.56 cm.).                            |
| Radiator hose bottom                             | Length 2½ in. (5.71 cm.).<br>Diameter 1½ in. (2.86 cm.).                            |
| First oversize bore                              | +0.020 in. (.50 mm.).<br>Actual bore 2.6381 in. (67 mm.).                           |
| Second oversize bore                             | +0.040 in. (1.00 mm.).<br>Actual bore 2.6581 in. (67.5 mm.).                        |
| Firing order                                     | 1-3-4-2.                                                                            |
| Piston clearance at pressure face below oil ring | .0021 in. to .0029 in. (.056 mm. to .071 mm.).                                      |
| Ring gap                                         | .006 in. to .010 in. (.152 mm. to .254 mm.).                                        |
| Number of compression rings                      | Two.                                                                                |
| Width of compression rings                       | 2.25 mm. (.0885 in.).                                                               |
| Number of oil rings                              | One.                                                                                |
| Width of oil ring                                | 4.0 mm. (.1575 in.).                                                                |
| Oil pressure relief valve operates               | 50 to 70 lb. per sq. in. (3.5 to 5.0 kg./cm.²).                                     |
| Oil pressure (normal)                            | 40 to 45 lb. per sq. in. (2.8 to 3.2 kg./cm.²).                                     |
| Gudgeon pin type                                 | Clamped.                                                                            |
| Gudgeon pin diameter                             | .7087 in. { +.0004 in.<br>+.0006 in.<br>18 mm. { +.010 mm.<br>+.015 mm.             |
| Fit in piston                                    | Double thumb press.                                                                 |
| Fit in connecting rod                            | Clamped.                                                                            |
| Crankpin diameter (standard)                     | 1.772 in. (45 mm.).                                                                 |
| Crankpin minimum diameter for regrind            | 1.722 in. (43.75 mm.).                                                              |
| Connecting rod—length between centres            | 7.008 in. (178 mm.).                                                                |
| Connecting rod—type of bearing                   | Shimless, steel-backed, white-metal-lined.                                          |
| Connecting rod—side clearance                    | .004 in. to .006 in. (.10 mm. to .15 mm.).                                          |
| Connecting rod—diametrical clearance             | .0005 in. to .002 in. (.011 mm. to .056 mm.).                                       |
| Number of crankshaft bearings                    | Three.                                                                              |
| Type of main bearings                            | Shimless, steel-backed, white-metal-lined.                                          |
| Standard main journal diameter                   | 2.047 in. (52 mm.).                                                                 |
| Main journals first regrind diameter             | 2.027 in. (51.49 mm.).                                                              |
| Main journals second regrind diameter (minimum)  | 2.007 in. (50.98 mm.).                                                              |
| Main bearings—length                             | Front 1.496 in. (38 mm.).<br>Centre 1.496 in. (38 mm.).<br>Rear 1.575 in. (40 mm.). |
| Centre main bearing—end clearance                | .0014 in. to .0037 in. (.035 mm. to .095 mm.).                                      |
| Main bearings—diametrical clearance              | .0008 in. to .003 in. (.020 mm. to .075 mm.).                                       |
| Crankshaft—end thrust taken on                   | Centre bearing.                                                                     |
| Number of camshaft bearings                      | Three.                                                                              |
| Type of camshaft bearings                        | White metal (front).<br>Zinc alloy (rear and centre).                               |

## GENERAL DATA—continued

|                                                 |     |                                                     |                                                     |                                                                                                                     |
|-------------------------------------------------|-----|-----------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Camshaft—bearing clearance                      | ... | ...                                                 | ...                                                 | Front .0016 in. to .004 in. (.04 mm. to .10 mm.).<br>Rear and centre .0018 in. to .0037 in. (.045 mm. to .095 mm.). |
| Camshaft—end thrust taken on                    | ... | ...                                                 | ...                                                 | Front end.                                                                                                          |
| Camshaft—end float                              | ... | ...                                                 | ...                                                 | .005 in. to .013 in. (.125 mm. to .325 mm.).                                                                        |
| Camshaft—drive (type)                           | ... | ...                                                 | ...                                                 | Duplex chain, $\frac{3}{8}$ in. pitch, 60 pitches, endless.                                                         |
| Valve timing markings                           | ... | ...                                                 | ...                                                 | White chain links and "T" marks on wheels.                                                                          |
| Exhaust valve throat diameter                   | ... | ...                                                 | ...                                                 | 1.024 in. (26 mm.).                                                                                                 |
| Exhaust valve diameter                          | ... | ...                                                 | ...                                                 | Head 1.221 in. (31 mm.); Stem .315 in. (8 mm.).                                                                     |
| Inlet valve throat diameter                     | ... | ...                                                 | ...                                                 | 1.181 in. (30 mm.).                                                                                                 |
| Inlet valve diameter                            | ... | ...                                                 | ...                                                 | Head 1.299 in. (33 mm.); Stem .315 in. (8 mm.).                                                                     |
| Valve seat angle                                | ... | ...                                                 | ...                                                 | 30°.                                                                                                                |
| Tappet type                                     | ... | ...                                                 | ...                                                 | Hollow.                                                                                                             |
| Valve lift—inlet                                | ... | ...                                                 | ...                                                 | .315 in. (8 mm.).                                                                                                   |
| Valve lift—exhaust                              | ... | ...                                                 | ...                                                 | .315 in. (8 mm.).                                                                                                   |
| Inlet valve opens                               | ... | ...                                                 | ...                                                 | 11° before T.D.C. } From                                                                                            |
| Inlet valve closes                              | ... | ...                                                 | ...                                                 | 57° after B.D.C. } Engine                                                                                           |
| Exhaust valve opens                             | ... | ...                                                 | ...                                                 | 52° before B.D.C. } XPAG/TD/2                                                                                       |
| Exhaust valve closes                            | ... | ...                                                 | ...                                                 | 24° after T.D.C. } No. 24116                                                                                        |
| Valve spring pressure—shut                      | ... | Inner 31 lb. at 1.753 in. (14.06 kg. at 4.453 cm.). | Outer 62 lb. at 1.847 in. (28.12 kg. at 4.692 cm.). | Total 93 lb. (42.18 kg.).                                                                                           |
| Valve spring pressure—open                      | ... | Inner 43 lb. at 1.438 in. (19.5 kg. at 3.678 cm.).  | Outer 80 lb. at 1.532 in. (36.29 kg. at 3.89 cm.).  | Total 123 lb. (55.79 kg.).                                                                                          |
| Inlet and exhaust valve working clearance (hot) | ... | ...                                                 | ...                                                 | Engines up to No. XPAG/TD2/24115 :<br>.019 in. (.48 mm.).<br>Engines from No. XPAG/TD2/24116 :                      |
| Inlet and exhaust valve working clearance (hot) | ... | ...                                                 | ...                                                 | .012 in. (.30 mm.).                                                                                                 |
| Valve guides                                    | ... | ...                                                 | ...                                                 | Removable.                                                                                                          |

### FUEL SYSTEM

|                         |     |     |     |                                |
|-------------------------|-----|-----|-----|--------------------------------|
| Fuel tank reserve level | ... | ... | ... | Warning light on panel.        |
| Fuel delivery           | ... | ... | ... | S.U. electric pump.            |
| Carburettors            | ... | ... | ... | S.U. semi-down draught.        |
| Carburettor needles     | ... | ... | ... | Standard—ES. Weak—AP. Rich—EM. |

### CLUTCH

|                              |     |     |     |                                                                              |
|------------------------------|-----|-----|-----|------------------------------------------------------------------------------|
| Type (up to Engine No. 9407) | ... | ... | ... | Borg & Beck 7½ in. (18.4 cm.) diameter. Dry plate.                           |
| Type (from Engine No. 9408)  | ... | ... | ... | Borg & Beck 8 in. (20.3 cm.) diameter. Dry plate.                            |
| Facing                       | ... | ... | ... | Type RYZ on 7 in. diameter clutch.<br>Moulded type on 8 in. diameter clutch. |

### GEARBOX

|             |     |     |     |         |                        |              |              |              |
|-------------|-----|-----|-----|---------|------------------------|--------------|--------------|--------------|
| Synchromesh | ... | ... | ... | ...     | Second, third and top. |              |              |              |
|             |     |     |     |         | Gearbox                | Overall 8/41 | Overall 8/39 | Overall 9/41 |
| Ratios      | ... | ... | ... | Top     | 1.00                   | 5.125        | 4.875        | 4.555        |
|             |     |     |     | Third   | 1.385                  | 7.098        | 6.752        | 6.309        |
|             |     |     |     | Second  | 2.07                   | 10.609       | 10.09        | 9.429        |
|             |     |     |     | Bottom  | 3.5                    | 17.938       | 17.06        | 15.942       |
|             |     |     |     | Reverse | 3.5                    | 17.938       | 17.06        | 15.942       |

## GENERAL DATA—continued

### FRONT SUSPENSION AND STEERING

|                                 |                                                                     |
|---------------------------------|---------------------------------------------------------------------|
| Camber (static position) ... .. | Nil (tolerance $\pm 1^\circ$ ).                                     |
| Castor angle ... ..             | $2^\circ \pm \frac{1}{2}^\circ$ with side-members parallel to road. |
| Toe-in ... ..                   | Nil.                                                                |
| King-pin inclination ... ..     | $9^\circ$ to $10\frac{1}{2}^\circ$ full bump.                       |
| Track ... ..                    | Front $47\frac{3}{8}$ in. (1.203 m.). Rear 50 in. (1.27 m.).        |
| Turning circle ... ..           | 31 ft. 3 in. (9.525 m.).                                            |
| Wheelbase ... ..                | 7 ft. 10 in. (2.388 m.).                                            |
| Tyre size ... ..                | 5.50—15.                                                            |
| Tyre pressures ... ..           | 18 lb. per sq. in. (1.27 kg./cm. <sup>2</sup> ).                    |

### REAR AXLE

|                      |                                  |
|----------------------|----------------------------------|
| Type of axle ... ..  | Semi-floating.                   |
| Type of drive ... .. | Hypoid.                          |
| Ratio ... ..         | 8/41, 8/39 or 9/41.              |
| Adjustment ... ..    | By spacers and special fixtures. |

### BRAKES

|                          |                                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------|
| Type ... ..              | Lockheed hydraulic two leading shoe (front) (9 in. dia.).                                   |
| Type of linings ... ..   | Ferodo MR.19.                                                                               |
| Lining size—front ... .. | 8.75 in. $\times$ 1.5 in. $\times$ 1.87 in. (22.22 cm. $\times$ 3.81 cm. $\times$ 4.7 cm.). |
| Lining size—rear ... ..  | 8.75 in. $\times$ 1.5 in. $\times$ 1.87 in. (22.22 cm. $\times$ 3.81 cm. $\times$ 4.7 cm.). |
| Number of rivets ... ..  | Twelve per lining.                                                                          |

### SPRINGS (front)

|                                  |                                                             |
|----------------------------------|-------------------------------------------------------------|
| Type ... ..                      | Coil.                                                       |
| Free length ... ..               | 9.59 in. $\pm \frac{1}{16}$ in. (24.36 cm. $\pm$ 1.58 mm.). |
| Mean coil diameter ... ..        | 3.238 in. (8.24 cm.).                                       |
| Number of effective coils ... .. | $7\frac{1}{2}$ .                                            |
| Diameter of wire (ground) ... .. | .498 in. (1.27 cm.).                                        |
| Maximum deflection ... ..        | 4.24 in. (10.78 cm.).                                       |

### SPRINGS (rear)

|                            |                                                                                           |
|----------------------------|-------------------------------------------------------------------------------------------|
| Type ... ..                | Half-elliptic.                                                                            |
| Length ... ..              | 42 in. (106.7 cm.).                                                                       |
| Width ... ..               | $1\frac{1}{2}$ in. (3.81 cm.).                                                            |
| Number of leaves ... ..    | Seven.                                                                                    |
| Thickness of leaves ... .. | $\frac{7}{16}$ in. (5.56 mm.).                                                            |
| Camber (free) ... ..       | 4.1 in. (10.4 cm.).                                                                       |
| Working load ... ..        | 500 lb. at $\frac{1}{2}$ in. positive camber.<br>(226.8 kg. at 12.7 mm. positive camber.) |

### HYDRAULIC DAMPERS

|             |                       |
|-------------|-----------------------|
| Type ... .. | Girling or Armstrong. |
|-------------|-----------------------|

### ELECTRICAL

|                                                         |                                                                                   |
|---------------------------------------------------------|-----------------------------------------------------------------------------------|
| Distributor rotation ... ..                             | Counter-clockwise.                                                                |
| Automatic advance ... ..                                | $32^\circ$ on crankshaft.                                                         |
| Contact breaker gap (first type) ... ..                 | .010 in. to .012 in. (.25 mm. to .30 mm.).                                        |
| Contact breaker gap (second type) ... ..                | .014 in. to .016 in. (.36 mm. to .41 mm.).                                        |
| Spark plug (engines prior to No. XPAG/TD2/22735) ... .. | Champion L.10S (standard equipment) (14 mm.).<br>K.L.G. E.80.<br>Lodge HN or HNP. |

## GENERAL DATA—continued

|                                              |     |     |     |     |     |                                                                      |
|----------------------------------------------|-----|-----|-----|-----|-----|----------------------------------------------------------------------|
| Spark plug (engines from No. XPAG/TD2/22735) | ... | ... | ... | ... | ... | Champion NA.8.                                                       |
| Spark plug gap                               | ... | ... | ... | ... | ... | ·020 in. to ·022 in. (·50 mm. to ·56 mm.).                           |
| Ignition timing                              | ... | ... | ... | ... | ... | T.D.C. (full retard).                                                |
| Charging system                              | ... | ... | ... | ... | ... | C.V.C.                                                               |
| Battery                                      | ... | ... | ... | ... | ... | Lucas GTW9A, 51-amp. at 10 hours. 12-volt.<br>Positive earth return. |

### CAPACITIES

|                |     |     |     |     |     |                                                                                                            |
|----------------|-----|-----|-----|-----|-----|------------------------------------------------------------------------------------------------------------|
| Sump           | ... | ... | ... | ... | ... | 9 pints (10·8 U.S. pints ; 5·1 litres) ; from Engine No. 14948, 10½ pints (12·6 U.S. pints ; 5·96 litres). |
| Gearbox        | ... | ... | ... | ... | ... | 1½ pints (1·5 U.S. pints ; ·71 litre).                                                                     |
| Rear axle      | ... | ... | ... | ... | ... | 2½ pints (2·7 U.S. pints ; 1·3 litres).                                                                    |
| Cooling system | ... | ... | ... | ... | ... | 12 pints (14·4 U.S. pints ; 6·8 litres).                                                                   |
| Brake fluid    | ... | ... | ... | ... | ... | 1 pint (1·2 U.S. pints ; ·57 litre).                                                                       |
| Fuel tank      | ... | ... | ... | ... | ... | 12½ gallons (15 U.S. gallons ; 57 litres).                                                                 |

### GENERAL DIMENSIONS

|                                        |     |     |     |     |     |                                 |
|----------------------------------------|-----|-----|-----|-----|-----|---------------------------------|
| Length                                 | ... | ... | ... | ... | ... | 145 in. (3·683 m.).             |
| Width                                  | ... | ... | ... | ... | ... | 58½ in. (1·489 m.).             |
| Height                                 | ... | ... | ... | ... | ... | 53 in. (1·346 m.).              |
| Ground clearance                       | ... | ... | ... | ... | ... | 6 in. (15·24 cm.).              |
| Weight (unladen)                       | ... | ... | ... | ... | ... | 18½ cwt. (2,072 lb.) (941 kg.). |
| Weight (laden and with two passengers) | ... | ... | ... | ... | ... | 22 cwt. (2,464 lb.) (1120 kg.). |

### TORQUE SPANNER DATA

|                                                          |     |     |     |     |                            |
|----------------------------------------------------------|-----|-----|-----|-----|----------------------------|
| Cylinder head stud nuts                                  | ... | ... | ... | ... | 500 lb. in. (5·75 m./kg.). |
| Connecting rod big-end bolts<br>(to next split pin hole) | ... | ... | ... | ... | 320 lb. in. (3·7 m./kg.).  |
| Main bearing cap nuts<br>(to next split pin hole)        | ... | ... | ... | ... | 750 lb. in. (8·6 m./kg.).  |
| Steering wheel attachment nut                            | ... | ... | ... | ... | 500 lb. in. (5·75 m./kg.). |
| Gudgeon pin clamp screw                                  | ... | ... | ... | ... | 400 lb. in. (4·6 m./kg.).  |